

Active Inference Math Session ~ July 19, 2024

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Duration: 31:10 | **Uploaded:** Unknown **Transcript method:** auto_caption

place where I was ready to take calculus and I also came back to uh higher education as a non-traditional student after about four or five years away from high school after I worked as a journalist for a few years so I I I say all this to say that when people think of people who are good at math they might think of someone who did like a who was in math contests or could calculate things quickly and I was not any of those I would say that one of the things I do well is working with mental models of things and looking at some different mental models of math has helped me a lot there are two mental models that I always find helpful to share at the outset the first is called the rule of four which is a way of thinking about how we represent mathematical Concepts and it's not just four it it could be more than four but the idea behind the rule of four which to my knowledge um was originated by uh Deborah Hughes Hallet and her co-authors in one of their calculus books is that there are multiple ways to represent mathematical ideas which are usually set up as graphs numbers words and symbols and that the more of these you can use and Link together the greater likelihood you have of making your ideas understood and also I would say of effectively solving problems later so for the sake of efficiency lots of times things are highly symbolic and highly compressed but the more we can decompress kind of unzip those equations and expand them out into experiential understanding and multiple representations that usually seems to add a lot of value to people's experience with math and one of the most destructive things that sometimes happens to people is they are pressured to compute quickly in a way that makes their selfworth seem as if it is at risk if they cannot do so and is usually what turns people off from math most people can remember the moment where they or I would say many people can remember the moment where they stopped liking math and so if we can just go back to before that time and work around it it seems to work pretty well the other major concept that I like to use which is not mass specific is well let me stop there is there anything y'all have any thoughts on that I agree with uh a lot of people get turned off of math based on personal experiences maybe with an instructor or just hitting a wall with a Concepts or not being fast enough things like that I can recall so many kids in school who are like very intelligent but then they had a subject like it

was math or chemistry and then the way the instructor taught and their reaction to their struggles just turn them off from the subject completely and we don't have to do that that's like the we're not on that path that has an objective like a traditional school is so we can look at all the different ways we could approach this terrain and each find our own part of it and how we Traverse it which is what's so exciting to me about being a mathematician and a pedagogist professor whatever outside of that setting where it's in a linear objective oriented way the other cognitive tool that I really find valuable as a high level thing is not math specific but it is about teaching but it's not it's useful in lots of other contexts outside of teaching this this idea of blooms taxonomy this is U something I used quite a bit when I was making exams or projects thinking about assessments that there are different levels of doing things and the types of things like using the multiplication table or taking a derivative is at the lowest level of understanding which is not to say that it's not important the lowest level can be foundational but it's not all there is I co-author I know I did not co-author let me speak accurately um I in toward a student who wrote an article about chain Rule and she was an artist so she was creating these images that were giving this very intuitive easy to understand way of working on this calculus concept and it was totally at the high level that is not hey can you apply this but it was about creating something that led to deeper understanding or more insight we don't have to go from the bottom up and we don't have to go we there's no specific entry point that you have to come at this from so if I were teaching this as my class which I am not I might ask students to create things well before they felt like they really understood them because you can do that uh you could ask someone just create an equation any equation make it as complex as you want and they don't necessarily have to be coming in with a a background of lots of practice before they do that but when I put these two things together what I get is an emphasis on multiple representations and multiple approaches that you have different representation systems verbal visual graphical numerical symbolic and you have multiple levels at which you can approach the activities and when you put that together there's just tons of different things you can do RJ I see you have your hand up please feel free to jump in I was just going to say on the um I I had done some work a long time ago uh in education technology and uh some of it was on um uh uh repetition based learning and that move you know that move from okay we're g to just get this as a rope you know you can remember this thing to actually understanding it uh uh there's something called The Generation effect which is is like you contributing to your model of the thing right that is like uh uh you know uh trying to visually represent the idea creating anything rather than just consuming uh it's it's it's you know like

they call it generation effect because has this this really really significant impact on on retention so even if you w trying to uh build on the thing but uh uh you know just just trying to um represent it in a different way has like this C crazy impact on your on your understanding of it and that I it's a um I think it's I think it's like the explanation for why it is like physical note-taking um has a a greater impact than um say like listening to a recorded lecture like you know you recorded the lecture in class and then afterward you listen to it it's very different from uh uh in terms of retention impact than than writing the notes as you go along um so it's kind of it's interesting because like you know taxonomy represented is like this linear up and down but you can kind of like go from jump to create and then come back down to evaluate and analyze anyway it's just it made me think of that yeah I really like that and I mean it's how often are people able to apply things before they understand them and they understand them through app right it's it's not like you have to jump in at one specific place and go up or down it's that there are multiple different things you can do so I kind of think of these like a pyramid I put the rule of four at the base uh and then I I stack these sliced triangles on top of it and uh end up with a mental model of a mental model of a cognitive space that you can navigate as a way of working with mathematics and so that might be helpful context for anyone watching this or coming to these meetings to understand why we're it might feel like we're jumping around but we're trying to map in lots of different ways and create connections between these different things I hope that makes sense and that might give some context for looking at so what I did in anticipation of this meeting um because I am not I would not claim the title of expert about active inference I've been familiar with it uh for about no I've been exposed to it for a little over a year but I'm just now getting to a place where I have time to really focus on it so I'm reading the textbook for the first time and what I did was look through the chapter two and through the appendices to see what might be um helpful so maybe some of these things with a little bit of priming when we see them if you haven't totally fine if you haven't read them yet um RJ knows I like a very improvisational person so when I was asked what should we do before tomorrow and was thinking that's a really it'd be great if I thought about that wouldn't it um the first thing I'm going to try to do is find the textbook which I think I found a moment ago but what I did was I went through the chapter two and the appendices to see both what the recommended mathematical background is and what the um what is coming up like next week in terms of what we might be exposed to so I thought we could walk through that a little bit with a particular emphasis on not here it is not the active inference per se but what ideas are we going to see and what what do this ideas mean and um how might we be

somewhat familiar with them so what I saw in chapter 2 just to reality check myself chapter 2 is the next thing that the group looks at right or is it two chapters per or two weeks per chapter it's it's if someone is able to check that would be great if not super it's not it's not a blocker um anyway let's let's talk about this which is the first math equation that pops up in the in the textbook and it's two fundamental rules of probability because probability is very important in this chapter as we talk about what is the certainty or the likelihood that something is true given the available evidence that we have so this first rule of probability quickly um to review what I always start with if I teach probability from foundations is probability is a way of assigning numbers to things that could happen that give us an intuitive sense of How likely or unlikely something is and probably anyone who comes to this these sessions has at least some intuitive idea of probability even if whether or not there's a formal definition and many people also have formal definitions um this first rule the sum of all events probability sum of the probability of all events has to equal one tells us something has to happen and this concept the name for this is the suu think but something has to happen so we have a probability space and we can divide that probability space which is everything that could happen we can divide that into events and then each event has a probability and if you take each of those distinct events and add them up it has to be one has to be 100% because something has to happen the other key observation that you can build off of is What's called the product rule um the notation there's a lot of different types of notation that pop up here so this probability of X, Y is called The Joint distribution and this is the or the joint probability excuse me and this is the probability that both X and Y happen you will see other notation for this this could also be the probability of X and Y some people like to use words instead of the comma um or you might also see this as the probability some older texts use a symbol that looks like this and I just realized how awful that looks when I type directly on white so I'm going to move this over here so there's the joint probability which is the probability that we're interested in two things and that this is the probability both of them happen there's the marginal probability which is the probability of one single event happen and there's the conditional probability so the marginal probability is just the probability of a single event The Joint probability is the probability of what's sometimes called a compound event where two things are happening and then there's the conditional probability which is the probability that something happens given a specific piece of [Music] information when you see p of XY or P of X or P of Y with this divider between it this is pronounced probability of Y given X which means I already know for sure that X has happened and it's not hard for us to think of specific cases where if I know

something has happened it drastically changes my probability uh of something else happened so these are the three different types of probability that show up in these rules I think it's worth talking about them because especially for people who aren't as you know deep in this you're going to hear these three different adjectives and it's good to recognize which is which the other thing is we're looking at this from a Bayesian perspective which means we think of probabilities as corresponding to beliefs so probability is not a fact about the universe a probability is a belief that we hold and that's sort of the justification for why if I know an additional piece of information it can change my belief about the probability of another piece of information being true I said a lot there and I'm not I don't want to lecture the entire time but does what I've said make sense here I think these are three concept steps worth highlighting cool so we can look at resources about the difference between those things and from here there's these Basics when I say basic I'm not saying easy I'm saying fundamental these different ideas about what you can do with these from here you can build product rule the way that product rule and sum rule play together which is marginalization and you can also build various forms of Bayes rule so if you look up Bayes rule on Wikipedia which we might as well do you it'll look different ways there are different forms that can be written in they all have the same justification what I really like about probability is unlike calculus the you can start with the few fundamental axioms of probability and really develop it in a logical way without needing too much time too much space so just starting with the two rules the product rule and the sum rule you can get to this uh Bayes theorem which tells you a relationship between two different conditional probabilities and two different marginal probabilities already here you'll see the difference in notation $\cup$ for compound events like this definition of conditional probability is what the book is calling product rule and already there's they're using the the intersection sign $\cap$ um how is this useful so I mean there's the formula you can practice using the formula you can look at the visual proof or intuition about what it means how is this useful this is useful the the classic example is about testing for a disease so the classic example here would be that you take a test for let's say cancer it's a popular example and it comes back positive you know the the false positive rate there is a false positive rate so you know the the false positive and false negative rates of the tests what is the probability that you actually do have cancer given that the test came back positive this is a classic example of how to apply Bayes rule um I'm sure I can find an example online where they go through it in detail and this is the type of calculation that shows up several times in this chapter at least conceptually where we observe something which should be the test being

positive and then we want to update our information about the likelihood of something which would be the disease the way that this occurs is um so that's the big picture idea of what this uh box is getting at and why it might be useful there's this terminology of Prior and likelihood and I like this example in the book of trying to determine whether you have an apple or a frog the prior here is the the prior belief that you have before you you know you get additional information so the prior would be before you take the T T How likely you think it is that you have the disease and then the likelihood model is something it would be about the test it would be what you know about the relationship between the test results and actually having the disease and I think the the book does a fairly good job of illustrating this maybe it would be worth us doing some exercises or something at some point but just for this high level overview that we're looking at and trying to maybe make it more comfortable let's skip like calculations I'll give you homework if you want it but I don't want to just start there and all of these calculations are using the rules in the box so being familiar with the rules in the Box are helpful to understand where these numbers actually come from the explanation is is great um then there's a notion of surprise surprise I I wasn't familiar with this unit but after I read about it it makes sense gats are like bits they're the sort of continuous version of bits bits measure how much information bits are a measurement of information that are expressed base two if if you if you're familiar all with that because either zero or one the discret binary and Nats are the analog they're just working base e so Nats are a different way of measuring how much information you get from something interesting I I I did not know that term gats and I feel like it's in in deep memory somewhere where where does that come from it comes from natural log oh okay I see all right and the same way that bit comes from U binary right um and the thing about is if you're working with A continuous probability distribution even if you're not um in like statistical thermodynamics there's uncertainty and it it may make more sense on some of the physical things to measure information in a continuous way and so the natur the natat actually plays a role analogous to the bit and you can convert between bits and NS just like you can convert between me meters and feet right they're just different units so gats are how you measure surprise and surprise is how much information you get from a specific observation and kind of a good place for us to wrap up is where do we get to like where we get to is that different models different probability distributions will have different different surprise statistics Associated to and okay let me let me back up I I'm there's a there's a common mistake here and I thought I would go ahead and make it actually a few different Notions of surprise that that measure different things and it's important to be careful about that

uh so one thing I should so using it's important to unpack two different Notions of surprise the first is the negative law evidence I don't want to get caught up in the details here but the idea is the this notion of surprise tells you how unlikely something is and there's a second notion called beian surprise which is actually about updating uh based on beliefs so in other words beian surprise quantifies the difference between a prior what we what we think How likely we think something is before we see and a posterior How likely we think something is after it actually happens and what that leads into is like the big math question which is if you wanted to quantify and say this probability distribution is like this one or it is not how would you do that and it turns out there are lots of different ways to do that and one of them is the cack liebler Divergence but the big picture on all of these things because I'm not going to have time to go deep in the details and and we it's not the purpose of this session anyway what all of these tools are doing is trying to give measurement to information and measurement to likelihood of of something happening or something being true and the symbols and everything else are just reporting that and giving concrete ways to calculate it and we can go deeper in the weeds if people want to but when we talk about these things we're talking about information and how much confidence we have something is true and what this does is give us tools to measure that cool so let's continue this in Discord I'm happy to continue it um but I need to stop the sync session now what I will say is I'm happy to give you homework if you want it if people want to get their